

Choosing the Bottleneck Dimension of Autoencoders and Variational Autoencoders from the Intrinsic Dimension of the Data: An ID-Guided Protocol with Active-Unit Verification

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Abstract

The bottleneck dimension m of autoencoders (AEs) and variational autoencoders (VAEs) is their most fundamental design parameter, yet in practice it is usually chosen by trial and error or convention. Under the manifold hypothesis, this paper proposes a conditional, diagnostic ID-guided protocol—a three-step procedure of estimation, setting, and verification—in which **a search window for m is derived from an estimate $\hat{d}_{ID}$ of the intrinsic dimension of the input data (AE: $m \in [\hat{d}_{ID}, 2\hat{d}_{ID}]$; VAE: $m \geq 2\hat{d}_{ID}$), and the active units (AU) of the trained VAE diagnose whether the surplus dimensions were actually truncated**. The rationale is the topological lower bound $m \geq d_{emb} \geq d_{ID}$ required for self-intersection-free reconstruction, and a water-filling-type activation condition $\lambda_j > \beta\tau^2$ in a linear-Gaussian surrogate. On synthetic manifolds with known ground truth and on fully connected VAEs on MNIST (3+ seeds each), the three steps operate under a reproducible decision rule (Swiss Roll: $AU_{sat} = d_{true} = 2$; Torus: $AU_{sat} = d_{emb} = 3$ for $m \geq 5$, with boundary over-pruning at $m \leq 4$ detected by the MSE cross-check; MNIST: $\hat{d}_{ID} = 10.03 \pm 0.12$ consistent with the AU slowdown region), and on the MNIST grid used in this paper the ID-guided search window reduces the training runs of a full grid search by roughly 60–80% while attaining the grid-best downstream linear-probe accuracy inside the window. We also quantify the limits of applicability: intrinsic-dimension estimation is stable only through reference representations with $m \gg \hat{d}_{ID}$ and inflates beyond $m \approx 2\hat{d}_{ID}$ on natural images, and for Conv-VAEs the AU truncation fails at $\beta = 4$, with β -strengthening restoring truncation but not order consistency (single-seed exploratory runs). The protocol is thus not a general solution: **$\hat{d}_{ID}$ provides a search window, and AU diagnoses the success or failure of truncation under a suitable β and architecture**.

Keywords: variational autoencoder; bottleneck dimension; intrinsic dimension; manifold hypothesis; active units; TwoNN; model selection; representation learning; knowledge extraction

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1. Introduction

1.1. Background: How Should the Bottleneck Dimension Be Chosen?

Autoencoders (AEs) and variational autoencoders (VAEs) [1] are core unsupervised models that learn low-dimensional latent representations from high-dimensional data [2]. Their most fundamental design parameter is the bottleneck dimension m : if m is too small, information is lost; if it is too large, redundant dimensions arise and, in VAEs, complicate

the interpretation of posterior collapse [3]. Nevertheless, in practice m is usually set to “a round number such as 32 or 64” or tuned by trial and error, and a selection principle grounded in the data itself has not been established.

On the other hand, according to the manifold hypothesis [4], natural data concentrate near a low-dimensional manifold inside the high-dimensional input space $\mathbb{R}^N$, and the local number of degrees of freedom is characterized by the intrinsic dimension $d_{\text{ID}} \ll N$. Indeed, estimators such as TwoNN [5] report that the intrinsic dimension of natural images is far smaller than the pixel dimension [7]. From this viewpoint, **the appropriate size of the bottleneck dimension m should be determined not by model-side convention but by a data-side quantity—the manifold dimension.** The question addressed in this paper is:

Can the bottleneck dimension m of an AE/VAE be chosen in a principled way from an estimate of the intrinsic (manifold) dimension of the input data? And is there a means of verifying, after training, that the choice was appropriate?

1.2. Central Idea and Contributions

The central claim of this paper can be stated in one sentence:

A search window for the bottleneck dimension m can be derived from an estimate $\hat{d}_{\text{ID}}$ of the intrinsic dimension of the input data as “AE: $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$; VAE: $m \geq 2\hat{d}_{\text{ID}}$,” and the active units (AU) of a VAE—the number of coordinates actually used under KL regularization—serve, under a suitable β and architecture, as a post-training diagnostic of whether the surplus dimensions were actually truncated.

This is not a claim of a general solution to choosing m ; it is the proposal of a **conditional, diagnostic workflow** (applicability conditions are summarized in Table 3). The claim is supported both theoretically and experimentally. On the theoretical side we use (i) the topological lower bound $m \geq d_{\text{emb}} \geq d_{\text{ID}}$ required to reconstruct a d -dimensional manifold without self-intersection (Proposition 1), and (ii) the activation condition $\lambda_j > \beta\tau^2$ obtained from a water-filling analysis of the linear-Gaussian setting (Proposition 2). The latter **predicts, within the linear-Gaussian surrogate**, that even if m is set above the required dimension, the KL regularization of the VAE truncates the surplus dimensions (posterior collapse) and the AU count reports the effective number of used dimensions—in other words, **whereas for an AE the choice of m itself fixes the representation dimension, for a VAE the safe-side operation “generous m + AU diagnosis” can be expected** (whether this expectation actually holds is itself checked by the Step-3 decision rule).

The contributions of this paper are threefold:

1. **An ID-guided bottleneck-dimension selection protocol** (Section 3): a diagnostic workflow consisting of (S1) estimating $\hat{d}_{\text{ID}}$ (TwoNN/MLE; for real data through a reference AE with $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$), (S2) setting m (AE: $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$ plus a topological margin; VAE: a generous setting $m \geq 2\hat{d}_{\text{ID}}$), and (S3) post-training verification by AU and reconstruction error, together with a **reproducible pass/fail decision rule** (Equations (2)–(3)), failure modes, and reporting requirements.
2. **Theoretical grounding of the selection principle** (Section 4): we organize the topological lower bound (Proposition 1) and the surrogate activation condition $\lambda_j > \beta\tau^2$ under a linear-Gaussian rate–distortion approximation (Proposition 2; τ^2 is the isotropic Gaussian observation-noise variance, and the correspondence between the KL term and mutual information is an approximation that neglects the aggregated-posterior mismatch—both stated in the claim body), and derive the operational implication that “the surrogate model predicts truncation of an excessive m , but the truncation threshold depends on β .”

3. **Multi-seed validation and quantified applicability boundaries** (Section 5): multi-seed validation of the full protocol is given on synthetic manifolds with known ground truth (the protocol returns d_{true} and d_{emb} under sufficient margin, with the Torus boundary over-pruning detected by the MSE cross-check; 3–5 seeds); on fully connected VAEs on MNIST, the AU verification is multi-seed (3 seeds; $\hat{d}_{\text{ID}} \approx 10$ –12 consistent with the AU slowdown region), with Step-1 stability supported by subsampling confidence intervals, cross-checks with estimators of different principles (DANCo, ESS), and the 3-seed reference-model comparison (Section 5.5). We also show with a linear probe that the ID-guided range reduces search cost while retaining downstream classification accuracy (Section 5.4). We further quantify the limits of applicability: (a) the estimation of $\hat{d}_{\text{ID}}$ is stable only through reference representations with $m \gg \hat{d}_{\text{ID}}$, while isometry improvements are secondary (four-model comparison on MNIST and Fashion-MNIST, 3 seeds; conditional-stability sketch in Proposition 3); (b) on natural images the TwoNN estimate agrees with reported values [7] up to $m \approx 2\hat{d}_{\text{ID}}$ but inflates beyond; (c) for Conv-VAEs at $\beta = 4$ the truncation threshold $d^*(\beta)$ exceeds 512, so AU verification requires a strengthened β . The Step-3 evidence for Fashion-MNIST, the β -strengthened Conv-VAE recovery (V1 only, not V2), and the natural-image AU plateau are single-seed exploratory observations (Appendix C), and Step 3 on natural images remains unresolved—the contributions are claimed inclusive of these limitations.

Relation to prior work: studies that “measure” the intrinsic dimension of deep representations [7–9] stop at analyzing trained networks, whereas this work uses intrinsic-dimension estimation for the **design of the bottleneck dimension** and formalizes a closed-loop verification of that design through AU. Whereas the linear analysis of posterior collapse [3] remains qualitative, we organize it into the closed-form threshold $\lambda_j > \beta\tau^2$ (the novelty lies in the integrated organization; Section 4) and operate it as the rationale of the selection protocol.

1.3. Two Faces of the “Manifold Dimension”: d_{ID} and d_{emb}

As the “manifold dimension of the input data” we distinguish two quantities: the **local intrinsic dimension** d_{ID} (the local degrees of freedom of the manifold; the target of TwoNN/MLE) and the **topological minimum embedding dimension** d_{emb} (the minimum dimension for embedding into Euclidean space without self-intersection). For the Swiss Roll $d_{\text{emb}} = d_{\text{ID}} = 2$, but for the Torus $d_{\text{emb}} = 3 > d_{\text{ID}} = 2$. What the bottleneck requires is d_{emb} (Proposition 1); since only d_{ID} can be estimated for real data, this gap (topological obstruction) is one reason for the margin “set m above $\hat{d}_{\text{ID}}$ ” (Section 3.2.2).

1.4. Limitations of This Paper (Summary)

Our validation is restricted to synthetic manifolds with known ground truth ($d_{\text{true}} \leq 3$), MNIST and Fashion-MNIST (fully connected AE/VAE, 3 seeds), and—as quantification of the applicability boundary—Conv-VAEs and natural images (CIFAR-10, SVHN). An estimate $\hat{d}_{\text{ID}}$ (TwoNN/MLE) that is “numerically stable” is not necessarily “unbiased with respect to the true intrinsic dimension” (stability vs. validity; Section 6.2); this paper treats $\hat{d}_{\text{ID}}$ as a **reference value** for selection and does not claim to estimate the true value. Automatic detection of change points (knees) on the AU curve is strongly grid- and seed-dependent, so it is not a named output of the protocol and is placed as an exploratory analysis in Appendix B. Full application to large-scale natural images remains future work.

The remainder of the paper covers related work (Section 2), the protocol (Section 3), its theoretical grounding (Section 4), experiments (Section 5), discussion (Section 6), and

conclusions (Section 7). Proofs, sensitivity analyses, and exploratory experiments are collected in the appendices.

2. Related Work

Representation learning and autoencoders: Bengio et al. [2] organized the framework in which AEs learn the low-dimensional structure of high-dimensional data based on the manifold hypothesis [4]. The β -VAE [11] controls the degree of compression of the representation through the KL weight β , which directly affects AU. DAE [15] and CAE [16] are regularized variants that promote learning of manifold structure; in this paper they serve as candidate reference representations for intrinsic-dimension estimation (Section 5.5).

Intrinsic-dimension analysis of deep representations: Pope et al. [7] showed that the intrinsic dimension of natural images is far lower than the pixel dimension (CIFAR-10: $d_{\text{ID}} \approx 26$), and Ansuini et al. [8] quantified how the ID changes across layers of deep networks. Valeriani et al. [9] analyzed the hidden-representation geometry of large transformers, and Birdal et al. [10] related ID to generalization. These are studies that “measure” trained models; our work is complementary in that it uses the measurements for the **design and verification of the bottleneck dimension**. As ID estimators we adopt TwoNN [5] and MLE [6] (DANCo [19] is more precise but computationally expensive and left as future work).

Active units and posterior collapse: AU [3,11] is the number of latent dimensions whose posterior-mean variance exceeds a threshold δ , and measures the number of coordinates actually used under KL regularization. Lucas et al. [3] analyzed the mechanism of posterior collapse in linear VAEs. This work incorporates that mechanism into the selection protocol as a “safety device that can truncate an over-generous m ” (whether truncation actually occurred is diagnosed in Step 3), with the water-filling activation condition as its theoretical rationale (Proposition 2; the PPCA of Tipping & Bishop [12] is the special case $\beta = 1$).

Automatic determination of the number of latent dimensions: within linear and probabilistic models, Bayesian PCA with an ARD (automatic relevance determination) prior [13] determines the number of principal components, and Bayesian nonparametric latent variable models such as those based on the Indian Buffet Process [14] determine the number of latent features, during learning. The KL truncation of the β -VAE performs a similar “automatic pruning of surplus dimensions” in the nonlinear setting, but the pruning result depends on β and the architecture (Section 5.6). Rather than relying on model-side automatic determination, our protocol derives a search window from the data-geometry side ($\hat{d}_{\text{ID}}$) and verifies it against the pruning result (AU); the two approaches are complementary.

Geometric autoencoders: Nazari et al. [18] and Kato et al. [17] proposed regularizers that preserve the geometry (isometry) of the latent space. We compare the effect of isometry on the stability of ID estimation against the “bottleneck margin $m/\hat{d}_{\text{ID}}$ ” and show that the latter is dominant (Section 5.5).

3. ID-Guided Protocol for Choosing the Bottleneck Dimension

3.1. Problem Setting and Main Symbols

We define an AE/VAE with input dimension N , bottleneck dimension m , encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$, and decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$. In a VAE the encoder outputs a posterior $q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$, and the KL regularization (strength β) acts as an information bottleneck.

We define the **active units (AU)** as

$$\text{AU}(m) = \sum_{k=1}^m \mathbf{1}[\text{Var}_x[\mu_k(x)] > \delta], \quad \delta = 10^{-2},$$

(1)

where $\mu_k(x)$ is the k -th component of the posterior mean; the threshold δ follows Lucas et al. [3], and we have confirmed that AU values are essentially unchanged for $\delta \in [10^{-3}, 10^{-1}]$ (Appendix D). AU is computed on test data (the train/test distinction is stated in each experiment section).

Table 1. Main symbols. Each quantity is distinguished so that the protocol’s claims are stated without conflating targets and estimators.

Symbol	Meaning	Role in the protocol
m	Bottleneck dimension	Design parameter to be chosen
d_{ID}	Intrinsic dimension of the data manifold	Target quantity (unknown for real data)
d_{emb}	Topological minimum embedding dimension ($\geq d_{\text{ID}}$)	Lower bound for m (Prop. 1)
$\hat{d}_{\text{ID}}$	TwoNN/MLE estimate of d_{ID}	Reference value for Step 1 (may be biased)
$\text{AU}(m)$	Number of KL-active coordinates, Equation (1)	Verification signal for Step 3
AU_{sat}	Saturated (plateau) AU value	Read-out of effective dimension (VAE)
$d^*(\beta)$	Water-filling threshold count (Prop. 2)	Idealized prediction of AU_{sat}
κ	Decoder Jacobian condition number	Isometry (distortion) indicator

Two distinctions deserve emphasis here. First, $\hat{d}_{\text{ID}}$ is a **reference value, not the true value**: even when TwoNN/MLE is numerically stable (low variance), it may carry a systematic bias due to geometric distortion of the reference representation (stability vs. validity; Section 6.2). Second, AU is **not an estimator of the intrinsic dimension**: AU is an observable—“the number of coordinates the KL regularization allowed to be used”—and depends on β , δ , and the architecture. The protocol is designed to avoid over-reliance on any single indicator by cross-checking these two quantities of different natures.

3.2. The Protocol

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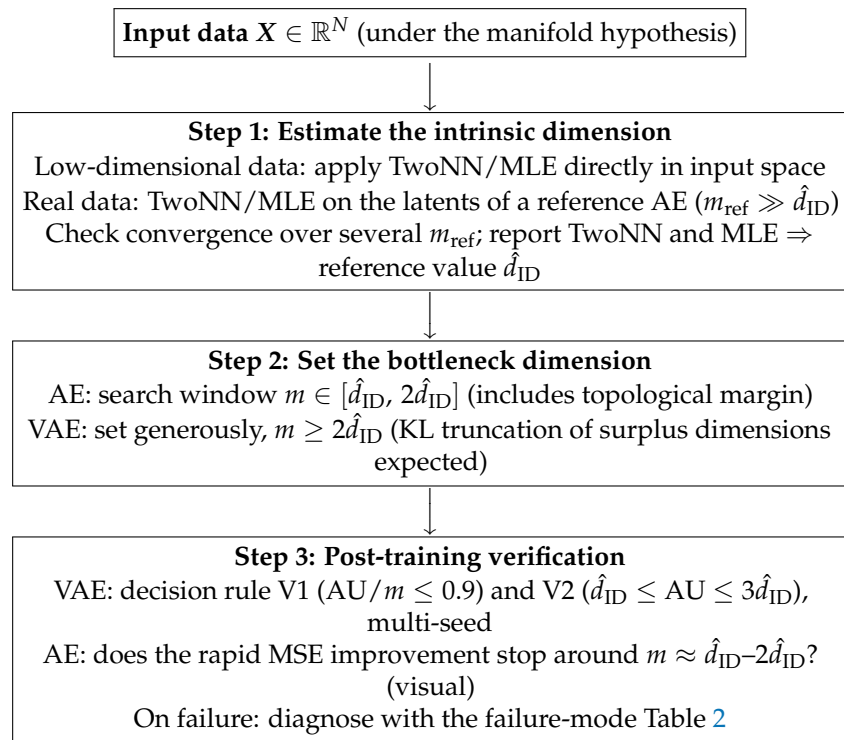


Figure 1. ID-guided protocol for choosing the bottleneck dimension m . The protocol closes the loop: the data-side estimate $\hat{d}_{\text{ID}}$ determines the design, and the model-side observable AU verifies the design after training.

3.2.1. Step 1: Estimating the Intrinsic Dimension

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For synthetic or low-dimensional data, TwoNN/MLE can be applied directly in input space. For high-dimensional real data (e.g., $N = 784$ for MNIST), direct estimation becomes unstable due to distance concentration, so we estimate **through the latent representation of a generous reference AE** ($m_{\text{ref}} \gg \hat{d}_{\text{ID}}$; $m_{\text{ref}} = 256$ in this paper). Two requirements are mandatory: (i) confirm convergence of the estimate over several m_{ref} (e.g., $\{64, 128, 256\}$); (ii) report two estimators with different principles (TwoNN and MLE) and confirm they agree in order of magnitude. For setting the window and the verification point in Steps 2–3, the **pre-registered primary estimator is the converged TwoNN value rounded to the nearest integer**; MLE (and, where computed, DANCo/ESS) serve as cross-checks (how a diverging cross-check is handled is specified in Step 3). That the condition $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ is the dominant condition for stable estimation is consistent with the conditional-stability sketch of Proposition 3 and supported by the experiments (Section 5.5). Conversely, $m_{\text{ref}} \lesssim 3\hat{d}_{\text{ID}}$ causes systematic underestimation, and on natural images inflation occurs for $m_{\text{ref}} \gg 2\hat{d}_{\text{ID}}$ (Section 5.6)—note that the trust region of Step 1 is bounded in m_{ref} .

3.2.2. Step 2: Setting the Bottleneck Dimension

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For AEs: for $m < d_{\text{emb}}$, self-intersection-free reconstruction is impossible in principle (Proposition 1), and any excess in m remains as redundant dimensions (an AE has no automatic truncation). We therefore take $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$ as the **default search window**, absorbing the possibility of a topological obstruction ($d_{\text{emb}} > d_{\text{ID}}$; Torus type) into the margin. Choose the upper side of the window if reconstruction accuracy matters, the lower side if compression matters. The factor 2 is not derived from first principles but is a default that worked in our experiments; topology, noise, class mixtures, or reconstruction-fidelity demands may require a larger margin (on the Torus, activation stabilizes only

at $m \geq 5 > 2\hat{d}_{\text{ID}} = 4$; Section 5.2)—the window is a starting point for the search, not a guaranteed upper bound.

For VAEs: Proposition 2 predicts, within the surrogate model, that the KL regularization truncates surplus dimensions even when m exceeds the required dimension, so **setting $m \geq 2\hat{d}_{\text{ID}}$ generously is the safe-side operation** (whether truncation actually occurred is checked by V1 in Step 3). The truncation threshold, however, depends on β ($d^*(\beta) = |\{j : \lambda_j > \beta\tau^2\}|$): if β is too small the truncation fails and AU sticks at m (detectable in Step 3).

3.2.3. Step 3: Post-Training Verification

After training, the design is verified with two signals (both over multiple seeds).

VAE: AU verification (with a pass/fail decision rule). The verification point m_{ver} is **pre-specified as the smallest tested grid point $\geq 2\hat{d}_{\text{ID}}$ —with $\hat{d}_{\text{ID}}$ the primary TwoNN estimate of Step 1—before the AU curve is observed** (to avoid post-hoc selection). When a cross-check estimator (e.g., MLE) yields a $2\hat{d}_{\text{ID}}$ that falls past the next grid point, verify there as well (on MNIST, MLE ≈ 11.9 points to $m = 32$ in addition to $m = 20$, and both pass; Section 5.3). For a VAE trained at m_{ver} , evaluate the following two conditions on the multi-seed mean AU:

$$\text{V1 (truncation occurred)} : \text{AU}(m_{\text{ver}})/m_{\text{ver}} \leq 0.9, \quad (2)$$

$$\text{V2 (order consistency)} : \hat{d}_{\text{ID}} \leq \text{AU}(m_{\text{ver}}) \leq 3\hat{d}_{\text{ID}}. \quad (3)$$

The verification **passes** when both hold (evidence of a compression to the same order as the manifold dimension of the data); failure of V1 suggests “ β too small relative to the decoder capacity,” failure of the upper side of V2 suggests “Step-1 underestimation or insufficient truncation,” and failure of the lower side suggests “over-regularization (β too large)” (Table 2). V1/V2 is **a conservative reporting convention, not a validated statistical test**: the threshold 0.9 and factor 3 are working defaults. In all settings of this paper, the verdicts on the multi-seed mean AU are invariant for a V1 threshold in (0.67, 0.94) and a V2 factor in (2.3, 6.6); recomputed per seed, the verdicts are unanimous at the default thresholds and invariant for a V1 threshold in (0.70, 0.93) and a V2 factor in (2.3, 6.6)—the narrower per-seed lower edge is set by one MNIST seed with AU = 14 at $m_{\text{ver}} = 20$ (Section 5; the supplementary analysis script recomputes both ranges; the displayed V2 ranges are rounded conservatively—the computed mean-verdict lower edge is 2.24). The width of the V2 band $[\hat{d}_{\text{ID}}, 3\hat{d}_{\text{ID}}]$ is a consequence of AU being a β - and architecture-dependent observable rather than an ID estimator (Section 3.1); V2 checks order consistency only, not that AU equals any dimension quantity—in particular, a pass does not imply $\text{AU}_{\text{sat}} = d_{\text{emb}}$ (see the Torus boundary behavior in Section 5.2). Simple manifolds show sharp saturation, while real data show a gradual slowdown (Section 5.3).

MSE verification (qualitative diagnostic; AE and VAE). Visually confirm that the rapid improvement of $\text{MSE}(m)$ slows down around $m \approx \hat{d}_{\text{ID}} - 2\hat{d}_{\text{ID}}$ (we do not propose an algorithmic elbow detector; automatic change-point detection is strongly grid- and seed-dependent and is confined to the exploratory analysis of Appendix B). For VAEs the MSE verification must **always be read together with V1/V2**: if AU passes but MSE keeps improving markedly beyond m_{ver} , suspect over-pruning due to a topological obstruction ($d_{\text{emb}} > d_{\text{ID}}$) and enlarge the margin (fourth row of Table 2; a concrete instance on the Torus is given in Section 5.2). The protocol is thus a **diagnostic workflow** that combines the mechanical V1/V2 verdict with the required plots and statistics (see the reporting requirements below); it is not a selection algorithm that outputs a single number.

Failure modes for a mismatched verification are organized in Table 2; this is a diagnostic checklist for readers applying the protocol to new data or architectures.

Table 2. Failure modes in Step 3 and their diagnoses. Each row is backed by the experiments in Section 5.

Symptom	Likely cause	Remedy / interpretation
$AU = m$ for all tested m (no truncation)	β too small relative to decoder power ($d^*(\beta) \gg m$)	Increase β or restrict decoder capacity; see Conv-VAE boundary (Section 5.6)
AU saturates far above $\hat{d}_{ID}$	Step-1 underestimate (m_{ref} too small), or entangled multi-class structure	Re-check Step-1 convergence in m_{ref} ; treat AU_{sat} as an upper reference
$\hat{d}_{ID}$ inflates as m_{ref} grows (no convergence)	Latent-geometry distortion at large m_{ref} (natural images)	Restrict to $m_{ref} \approx 2\hat{d}_{ID}$ and cross-check with the input-space estimate
MSE keeps improving beyond $2\hat{d}_{ID}$	Topological obstruction ($d_{emb} > d_{ID}$) or noise fitting	Allow a larger margin; inspect reconstruction quality

Reporting requirements: when reporting an application of this protocol, state at minimum (i) the number of seeds and the inter-seed variance of $\hat{d}_{ID}$ and AU, (ii) the m -grid and the values of m_{ref} , (iii) β , δ , and preprocessing, and (iv) both the TwoNN and MLE estimates. The main experiments of this paper report these quantities where applicable; for the boundary and exploratory experiments the complete set (including MLE and inter-seed variances) is contained in the supplementary logs.

3.3. Applicability (Summary in Advance)

Table 3 summarizes in advance where the protocol works and where it reaches its boundary (evidence in Section 5). Readers can first check whether their setting is within the applicable range.

Table 3. Applicability of the ID-guided protocol (summary of the evidence in Section 5). “Stable” means low inter-seed variance under our experimental conditions, not unbiasedness.

Setting	Step 1 ($\hat{d}_{ID}$)	Step 3 (AU verification)	Overall
Synthetic manifolds ($d_{true} \leq 3$, 3–5 seeds)	Direct input-space estimate; matches d_{true}	Pass: saturation at d_{true} / d_{emb} with sufficient margin ($\sigma = 0$); Torus boundary over-pruning ($m \leq 4$) caught by the MSE cross-check	Fully applicable (ground-truth validated)
FC-VAE on MNIST (3 seeds) / Fashion-MNIST (Step 1: 3 seeds; Step 3: single seed)	Stable: $10.03 \pm 0.12 / \approx 10$	Pass (V1/V2) on MNIST; Fashion-MNIST exploratory	Applicable (MNIST fully validated)
Conv-VAE on MNIST	Stable (inflates while $AU = m$)	Fail at $\beta_{max} = 4$ (V1); $\beta_{max} \geq 10$ restores V1 but not V2 (single seed)	Applicable only with strong β (exploratory)
Natural images (CIFAR-10 / SVHN, 3 seeds)	CIFAR-10 consistent with Pope et al. [7] at $m_{ref} \approx 2\hat{d}_{ID}$; inflates beyond	Unresolved: no V1/V2 pass within tested range	Step 1 usable with care; Step 3 open problem

4. Theoretical Grounding of the Selection Principle

We state the status of the results up front. Proposition 1 is a rigorous lower bound under an idealization (zero reconstruction error); Proposition 2 is an analytical result **restricted to the linear-Gaussian setting**, and its application to nonlinear VAEs is a qualitative guide (an interpretive surrogate); Proposition 3 is a sketch of conditional stability that grounds the design of Step 1.

4.1. Necessary Condition: Topological Lower Bound

Proposition 1 (Topological lower bound for the bottleneck dimension). *Let $\mathcal{M} \subset \mathbb{R}^N$ be a compact smooth d -dimensional manifold and let $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$ be continuous. If $g \circ f$ is the identity on $\mathcal{M}$ (zero reconstruction error), then $m \geq d_{\text{emb}} \geq d$, where d_{emb} is the topological minimum embedding dimension of $\mathcal{M}$.*

Proof. From $g(f(x)) = x$ for $x \in \mathcal{M}$, the restriction $f|_{\mathcal{M}}$ is a continuous injection. A continuous injection from a compact space into a Hausdorff space is a homeomorphism onto its image, so $f|_{\mathcal{M}}$ is a topological embedding; no topological embedding of a d -dimensional manifold into $\mathbb{R}^m$ with $m < d_{\text{emb}}$ exists by the definition of d_{emb} [24]. $\square$

The proposition is a necessary condition under the idealization of zero error and continuity; it does not directly apply to real neural AEs with finite error, noise, or non-injective behavior on small sets, and we use it only as an approximate guide for the lower end. This is the rationale for the lower end of Step 2. For real data d_{emb} is unknown, but since $d_{\text{emb}} \geq d_{\text{ID}}$, the estimate $\hat{d}_{\text{ID}}$ provides the reference point “below this is insufficient in principle,” and the possibility $d_{\text{emb}} > d_{\text{ID}}$ (a Torus-type topological obstruction) is one reason for the margin “set m above $\hat{d}_{\text{ID}}$.” The experiments confirm that this lower bound appears as the MSE elbow for the Swiss Roll ($d_{\text{emb}} = 2$) and the Torus ($d_{\text{emb}} = 3 > d_{\text{ID}} = 2$) (Section 5.2).

4.2. Mechanism on the Sufficient Side: Water-Filling Surrogate Activation Condition

Proposition 2 (Surrogate activation condition under a linear-Gaussian rate–distortion approximation). *Consider a β -VAE with linear encoder/decoder and Gaussian prior (details in Appendix A). Here τ^2 is the isotropic Gaussian observation-noise variance of the decoder output (the variance parameter of the likelihood $p(x|z) = \mathcal{N}(g(z), \tau^2 I)$), and λ_j are the eigenvalues of the covariance matrix of the centered input data. We further use the rate–distortion approximation that replaces the KL term by mutual information (this replacement is an approximation that neglects the mismatch between the aggregated posterior and the prior; the validity of this proposition is limited to the range of this approximation). Then, from the water-filling theorem and the KKT conditions:*

1. (Activation threshold) The j -th eigendirection is active if and only if

$$\lambda_j > \beta \tau^2. \quad (4)$$

2. (Optimal number of active directions) The number of active eigendirections at the global optimum is $A^*(m) = \min(m, d^*(\beta))$ with $d^*(\beta) := |\{j : \lambda_j > \beta \tau^2\}|$.
3. (Monotone saturation) $A^*(m)$ is nondecreasing in m and saturates for $m \geq d^*(\beta)$.

This proposition is not a theorem about trained nonlinear VAEs but an **interpretive surrogate** for reading nonlinear experiments (derivation in Appendix A).

Remark 1 (Implications for the protocol). *All statements below are predictions within the surrogate model and are operated as diagnostic expectations for nonlinear VAEs (their success or failure is judged in Step 3). (i) Surplus dimensions with $m > d^*(\beta)$ are truncated at the optimum*

of the surrogate—this is the rationale for Step 2, “a generous m is safe for VAEs.” (ii) The truncation threshold $\beta\tau^2$ is proportional to β —if β is too small then $d^*(\beta) > m$ and no truncation occurs ($AU = m$). This is the failure mode in the first row of Table 2 and is actually observed for Conv-VAEs and natural images (Section 5.6). (iii) For $\beta = 0$ (vanilla AE) the threshold vanishes and all dimensions are used—the reason an AE has no automatic truncation and m itself must be constrained in Step 2.

Remark 2 (Surrogate rate vs. empirical AU). What Proposition 2 counts is the number of eigendirections with positive rate $R_j^* > 0$; what the empirical AU (Equation (1)) counts is the number of coordinates whose posterior-mean variance exceeds a threshold—by definition, two different notions. In the linear-Gaussian setting they coincide: at the optimum, $R_j^* > 0 \iff$ the posterior mean $\mu_j(x)$ depends on x ($\text{Var}_x[\mu_j] > 0$) $\iff$ AU counts the coordinate as active, while $R_j^* = 0 \iff \mu_j \equiv 0$ (collapse). For nonlinear VAEs this correspondence is only a **qualitative expectation**, and we do not claim that the surrogate predicts nonlinear AU counts quantitatively. In actual nonlinear VAEs, each m is optimized independently and stochastically, so the empirical $AU(m)$ carries non-monotone fluctuations, and the sharp saturation of the linear setting is replaced by a gradual slowdown (Section 5.3). Moreover, replacing the effective noise variance of a nonlinear decoder by a calibrated value τ_{eff}^2 “back-computed to match the observed effective dimension” (Section 5.6) is a **heuristic extrapolation**, not a consequence of this proposition.

4.3. Grounding of Step 1: Conditional Stability of Intrinsic-Dimension Estimation

Proposition 3 (Conditional stability of TwoNN (sketch)). If the encoder f is locally bi-Lipschitz (distortion $\varepsilon \in [0, 1)$) and neighbor ranks are preserved with high probability (Trustworthiness $\geq 1 - \delta_T$ with small δ_T), then the change in the distribution of the two-neighbor distance ratio r_2/r_1 is bounded by a continuous function $C(\varepsilon, \delta_T)$ (with $C \rightarrow 0$ as $\varepsilon, \delta_T \rightarrow 0$), and the deviation between the input-space and latent-space TwoNN estimates is governed by the same bound. This proposition is not a guarantee of unbiasedness but a sketch of sufficient conditions for conditional stability (proof sketch in Appendix A).

The experiments (Section 5.5) show that improving δ_T (neighbor-rank preservation) via $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ dominates over improving ε (isometry)—this is the design rationale for prioritizing bottleneck margin over isometric regularization in Step 1.

5. Experiments

The experiments are organized in three layers: (i) **principle validation**—does the protocol return the correct dimension on synthetic manifolds with known ground truth (Section 5.2)? (ii) **real-data validation**—does the full protocol work on MNIST and Fashion-MNIST, and does it have practical value in search cost and downstream performance (Sections 5.3–5.5)? (iii) **applicability boundary**—which conditions break for Conv-VAEs and natural images (Section 5.6)? **All main-text VAE verification results used for the principal verdicts are multi-seed experiments with at least 3 seeds (42, 123, 777);** two reused training sweeps are single-seed and are labeled as such where they appear (the Step-1 reference-AE sweep of Table 6, also re-analyzed in Tables 7 and 9, and the AE side of the downstream comparison); single-seed exploratory observations (the strengthened- β Conv-VAE runs and Fashion-MNIST Step 3) are labeled as such where summarized, with details in Appendix C.

5.1. Experimental Setup

Common: Adam ($\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, no weight decay), batch size 128, ReLU (Sigmoid output), PyTorch 2.0, a single NVIDIA GeForce RTX 3070 GPU (CPU fallback when CUDA is unavailable). Preprocessing is $[0, 1]$ pixel normalization only (synthetic manifolds are standardized to zero mean and unit variance). Train/test subsets

use the first $n_{\text{train}}/n_{\text{test}}$ samples of the official splits in fixed order. AU, TwoNN, and MLE are computed on the test subset.

Synthetic manifolds: hidden layers [64, 32], 200–300 epochs, VAE $\beta = 4.0$, $n = 5,000$ (4,000 train / 1,000 test).

MNIST / Fashion-MNIST: input dimension 784, hidden layers [512, 256], VAE $\beta = 4.0$, m -grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$ (changes per experiment are stated); MNIST: 300 epochs ($n_{\text{train}} = 20,000$, $n_{\text{test}} = 3,000$); Fashion-MNIST: 300 epochs (Exp-Real-Geom; $n_{\text{train}} = 20,000$).

ID estimation: TwoNN ($k = 2$) [5], MLE ($k = 10$; stability confirmed for $k \in \{5, 10, 20\}$) [6]. AU threshold $\delta = 10^{-2}$ (Appendix D).

Reproducibility and supplementary-file inventory: the supplementary material consists of four parts: (i) the experiment scripts (in one-to-one correspondence with the experiment names in Table A4); (ii) analysis scripts that regenerate every table and figure of the main text from the training logs; (iii) the training logs (JSON; all seeds); and (iv) environment information (random seeds, library versions). We verified before submission that the numbers in the tables match the output of the included scripts. The material is attached for review; upon acceptance it will be deposited in a public repository (GitHub/Zenodo) with URL and DOI.

5.2. Principle Validation: Synthetic Manifolds with Known Ground Truth

We validated the three protocol steps against ground truth on the Swiss Roll ($d_{\text{true}} = d_{\text{emb}} = 2$) and the Torus ($d_{\text{true}} = 2$, $d_{\text{emb}} = 3$) (3 seeds).

Step 1: the input-space TwoNN gives ≈ 2.0 for both the Swiss Roll and the Torus, matching d_{true} .

Step 3 (AE: MSE verification): as shown in Table 4, the rapid MSE improvement stops at $m = 2 = d_{\text{emb}}$ for the Swiss Roll and at $m = 3 = d_{\text{emb}}$ for the Torus—the lower bound of Proposition 1 appears as the actual MSE elbow, and for the Torus the topological obstruction “ $d_{\text{ID}} = 2$ is not enough” is observed. This is the experimental basis for the margin in Step 2 (set m above $\hat{d}_{\text{ID}}$).

Table 4. Synthetic manifolds (Exp-Syn-1): AE reconstruction MSE (mean $\pm$ std, 3 seeds). The elbow appears exactly at $m = d_{\text{emb}}$.

m	Swiss Roll MSE	Torus MSE
1	1.016 ± 0.315	0.356 ± 0.022
2	$0.029 \pm 0.001 \leftarrow \text{elbow } (= d_{\text{emb}})$	0.109 ± 0.007
3	$(6 \pm 2) \times 10^{-4}$	$(4 \pm 3) \times 10^{-4} \leftarrow \text{elbow } (= d_{\text{emb}})$
4	$(5 \pm 1) \times 10^{-4}$	$(1.0 \pm 0.4) \times 10^{-4}$

Step 3 (VAE: AU verification): as shown in Table 5, the AU of the VAE ($\beta = 4$) saturates at $\text{AU}_{\text{sat}} = 2 = d_{\text{true}}$ for the Swiss Roll over the whole tested range $m \in [2, 16]$, and at $\text{AU}_{\text{sat}} = 3 = d_{\text{emb}}$ for the Torus for $m \geq 5$, in all seeds (standard deviation 0.0; the activation at $m = 5$ is from the fine-grid experiment Exp-Syn-Verify). In contrast, the coordinate usage of the vanilla AE sticks to m (no truncation mechanism)—consistent with implications (i) and (iii) of Proposition 2: the surrogate prediction that *the VAE truncates the surplus dimensions of a generous m and AU reports the result* is confirmed on ground truth in this setting. However, in the boundary zone $m \in \{3, 4\}$ the Torus is over-pruned to $\text{AU} = 2 < d_{\text{emb}}$ —detailed next.

Table 5. Synthetic manifolds (Exp-Syn-AU): active units of the VAE ($\beta = 4$) vs. the coordinate usage of the vanilla AE (mean $\pm$ std, 3 seeds). “AU” for the AE is a pseudo-AU defined by the encoding-variance threshold (auxiliary observation).

	m	Vanilla AE	VAE ($\beta = 4$)
Swiss Roll ($d_{\text{true}} = d_{\text{emb}} = 2$)	2	2	2.0 ± 0.0 (saturation = d_{true})
	4	4	2.0 ± 0.0
	8	8	2.0 ± 0.0
	16	16	2.0 ± 0.0
Torus ($d_{\text{true}} = 2, d_{\text{emb}} = 3$)	4	4	2.0 ± 0.0 (over-pruned , $< d_{\text{emb}}$; see text)
	6	6	3.0 ± 0.0 (saturation = d_{emb})
	8	8	3.0 ± 0.0
	16	16	3.0 ± 0.0

Boundary-zone over-pruning and its detection by the MSE cross-check (Exp-Syn-Verify): for $m \in \{3, 4\}$ the Torus VAE settles at $\text{AU} = 2 < d_{\text{emb}} = 3$. This is neither a reporting error nor a fluke but **a genuine phenomenon that reproduces perfectly across seeds**: in a fine-grid confirmation experiment ($m \in \{3, 4, 5, 6, 8\}$, 5 seeds; Exp-Syn-Verify), all seeds give $\text{AU} = 2$ for $m \in \{3, 4\}$ and $\text{AU} = 3$ for $m \geq 5$ —the transition at $m = 5$ is deterministic. Over-pruning carries a clear reconstruction penalty: the MSE improves rapidly from 0.220 ± 0.002 at $m = 4$ to 0.138 ± 0.001 at $m = 5$ (where the third coordinate activates) and 0.073 ± 0.006 at $m = 6$, because a self-intersection-free embedding becomes possible. The interpretation is that under the KL pressure of $\beta = 4$, in the boundary zone with barely any surplus dimensions the reconstruction gain of keeping a third coordinate does not overcome its KL cost, and the model is pushed into a self-intersecting two-dimensional representation (an immersion)—**the activation of the topologically required dimension stabilizes across all seeds only for $m \geq 5$, just outside the upper end $2\hat{d}_{\text{ID}} = 4$ of the default search window**. This observation has two operational implications. First, it demonstrates on ground truth why the Step-2 margin is needed: on the Torus, the topological gap $d_{\text{emb}} > d_{\text{ID}}$ makes the default window $[\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}] = [2, 4]$ insufficient, and stability is reached only outside the window—the reason the window is a starting point, not a guaranteed upper bound (Section 3.2.2). Second, it **makes a limitation of V1/V2 explicit**: at the pre-specified verification point $m_{\text{ver}} = 4 = 2\hat{d}_{\text{ID}}$, $\text{AU} = 2$ passes both V1 ($r = 0.5$) and V2 ($2 \leq 2 \leq 6$)—because V1/V2 verifies order consistency with $\hat{d}_{\text{ID}}$, not that $\text{AU}_{\text{sat}} = d_{\text{emb}}$. What catches this miss is the accompanying MSE verification: the MSE keeps improving markedly beyond $m_{\text{ver}} = 4$ ($0.22 \rightarrow 0.07$), so the fourth row of Table 2 (topological obstruction) fires and an enlarged margin is prescribed.

Interpretive caution (distinguishing d_{ID} , d_{emb} , and AU): this result must not be read as “AU estimates d_{emb} .” What TwoNN/MLE estimate is the local intrinsic dimension d_{ID} ; what appears in the lower bound of Proposition 1 is the topological embedding dimension d_{emb} ; and AU is an observable “number of used coordinates” that depends on β , δ , and the architecture (Table 1)—three quantities with different definitions. The correspondence $\text{AU}_{\text{sat}} = d_{\text{emb}}$ on the Torus is a **conditional empirical observation**—under sufficient β , convergence, standardization preprocessing, and sufficient margin ($m \geq 5$ on the Torus), the number of coordinates remaining after KL truncation happened to coincide with the dimension required for embedding—not a theoretical necessity (generalization beyond the Torus and Möbius band is unconfirmed; single-seed topological-manifold experiments are in Appendix C).

Verdict (Step-3 rule; m_{ver} pre-specified as the smallest tested point $\geq 2\hat{d}_{\text{ID}}$): the Swiss Roll ($m_{\text{ver}} = 4 = 2\hat{d}_{\text{ID}}$: $\text{AU} = 2.0$, activation ratio 0.50) **passes** V1, V2, and the MSE verification (3 seeds, $\sigma = 0$). The Torus at $m_{\text{ver}} = 4$ passes V1 and V2, but **the MSE verification detects the topological obstruction** (a $3\times$ MSE improvement from $m = 4$ to

$m = 6, 0.220 \rightarrow 0.073$)—following the fourth row of Table 2, the margin is enlarged and all checks pass at $m = 6$ (**pass**; $AU_{\text{sat}} = 3 = d_{\text{emb}}$, stable for all larger m ; 3–5 seeds, $\sigma = 0$). This two-stage trajectory is itself a working example of the diagnostic workflow in which AU is always read together with MSE.

5.3. Real-Data Validation: MNIST

5.3.1. Step 1: Estimating $\hat{d}_{\text{ID}}$

We computed TwoNN/MLE on the latents of reference AEs (300 epochs; Table 6). TwoNN converges to around 10.0 for $m_{\text{ref}} \geq 64$ (the 3-seed value on the VAE-side latents is 10.03 ± 0.12 ; Table 8), and MLE (11.3–11.9) gives the same order. The Step-1 requirements—convergence over several m_{ref} and agreement of the two estimators—are met, yielding the **reference value** $\hat{d}_{\text{ID}} \approx 10$ –12 for **MNIST** (primary TwoNN value $\hat{d}_{\text{ID}} = 10$; MLE 11.3–11.9 as the cross-check; Section 3.2.1). Note that each row of Table 6 is itself an m -sweep of a plain AE, so its MSE column is also used for the Step-3 AE-side MSE verification below. As a contrast, linear PCA requires 43 dimensions for 80% cumulative variance and 86 for 90% (Figure 2, left)—cumulative variance dimensions are not intrinsic-dimension estimates, so this is an illustrative contrast rather than a direct baseline, but it shows that the dimensionality required for linear compression far exceeds the nonlinear estimate $\hat{d}_{\text{ID}} \approx 10$ –12.

Table 6. MNIST Step 1: TwoNN/MLE estimates through reference AEs of increasing m_{ref} . **All rows are a single-seed (42) exploratory sweep** (300 epochs); statistical uncertainty at $m_{\text{ref}} \in \{64, 128, 256\}$ is quantified separately by subsampling CIs and estimator cross-checks in Table 7 (Section 5.3.2), and the 3-seed reproducibility of the latent estimate is reported in Table 8.

m_{ref}	MSE	TwoNN ID	MLE ID	CKA
8	0.0191	6.08	6.70	0.725
16	0.0121	7.96	8.61	0.812
32	0.0087	8.84	9.89	0.875
64	0.0062	9.54	11.26	0.908
128	0.0061	9.84	11.45	0.921
256	0.0057	10.02	11.85	0.953

5.3.2. Statistical Robustness of Step 1: Subsampling CIs and Estimator Cross-Checks

Since $\hat{d}_{\text{ID}}$ provides the center of the search window, we quantify its statistical uncertainty (Exp-Boot-CI; a re-analysis of the latents of the trained reference AEs). Because bootstrap resampling with replacement corrupts kNN-based estimators through duplicated points, we report 95% percentile intervals from subsampling without replacement ($n = 1500, 200$ repeats). In addition, we apply DANCo [19] and ESS [20] (scikit-dimension implementations)—estimators of different principles—to the same latents and check order-of-magnitude agreement with TwoNN/MLE.

Table 7. Exp-Boot-CI (MNIST): statistical robustness of Step 1. Subsampling 95% intervals ($n = 1500, 200$ repeats, without replacement) and cross-check estimators (DANCo, ESS) on the same reference-AE latents (single-seed-42 models; the uncertainty quantified here is the sampling variability of the estimators).

Representation	TwoNN [95% CI]	MLE [95% CI]	DANCo	ESS
Raw input ($N = 784$)	12.63	14.46	784 (degenerate)	18.67
AE latents, $m_{\text{ref}} = 64$	9.56 [8.92, 10.24]	11.14 [10.91, 11.41]	10.04	13.17
AE latents, $m_{\text{ref}} = 128$	9.72 [9.22, 10.23]	11.33 [11.08, 11.56]	10.03	13.72
AE latents, $m_{\text{ref}} = 256$	9.93 [9.37, 10.68]	11.67 [11.45, 11.90]	10.95	14.98

The results (Table 7) can be summarized in three points. First, **the subsampling 95% intervals are narrow**—width 1.0–1.3 for TwoNN and 0.5 for MLE—so the read-out of the search-window center $\hat{d}_{\text{ID}} \approx 10$ –12 is robust to sampling variability. Second, **DANCo, an estimator of a different principle, agrees closely with TwoNN at 10.0–10.9**, and ESS is somewhat higher at 13–15 but of the same order—the Step-1 read-out is not an artifact peculiar to TwoNN behavior. In the raw input space, TwoNN (12.63) and MLE (14.46) deviate only moderately from the latent estimates, whereas DANCo degenerates to the ambient dimension 784 (a known failure mode of its angle/norm-statistics calibration when the ambient dimension is large relative to the sample size, not a software fault) and ESS inflates to 18.7. The reference-AE route is therefore **a pragmatic route that stabilizes the chosen estimators under our conditions**, not a uniquely validated one—some estimators give the same order even in raw input space. Third, the subsample-size sensitivity ($n = 500$ –2500; see the supplementary material) shows the TwoNN mean rising monotonically with n (e.g., $9.07 \rightarrow 9.72$ at $m_{\text{ref}} = 64$)—a direct illustration of this paper’s characterization of TwoNN as “numerically stable but with a sample-size-dependent bias” (Section 6.2); the subsample means sit slightly below the full-sample values of Table 6 for this reason.

5.3.3. Step 2: Setting the Search Window

From $\hat{d}_{\text{ID}} \approx 10$ –12, the search window is $m \in [10, 24]$ for the AE and a generous $m \geq 20$ for the VAE (the AE window endpoints use the TwoNN–MLE envelope, whereas the VAE verification point follows the primary-TwoNN rule of Section 3.2.3).

5.3.4. Step 3: Verification by AU and MSE (3 Seeds)

Table 8 shows $\text{AU}(m)$ of the VAE ($\beta = 4$, 300 epochs, 3 seeds). AU is highly reproducible across seeds ($\sigma \leq 0.5$); nearly all dimensions are active for $m \leq 12$ ($\text{AU} \approx m$), while the growth clearly slows down beyond $m > 12$, reaching only $\text{AU} = 22.3$ at $m = 64$ —the verification signal expected by Step 3: **KL truncation sets in near $\hat{d}_{\text{ID}} \approx 10$ –12**. On the AE side, the rapid MSE improvement also slows at $m \approx 16$ –20 (the MSE column of Table 6, which is the plain-AE m -sweep), consistent with the search window $[10, 24]$.

Verdict (Step-3 rule; $m_{\text{ver}} = 20$ pre-specified as the smallest tested grid point $\geq 2\hat{d}_{\text{ID}} = 20$ from the primary TwoNN estimate $\hat{d}_{\text{ID}} = 10$): $\text{AU} = 13.3 \pm 0.5$ with activation ratio $0.67 \leq 0.9$ (V1 holds) and $\hat{d}_{\text{ID}} = 10 \leq 13.3 \leq 30 = 3\hat{d}_{\text{ID}}$ (V2 holds): **pass** (per-seed verdicts agree 3/3). The verdict also passes at the MLE-shifted verification point $m = 32$ (MLE ≈ 11.9 gives $2\hat{d}_{\text{ID}} \approx 24$, whose next grid point is 32; Section 3.2.3) and at $m = 64$ (activation ratios 0.49 and 0.35; $\text{AU} = 15.7$ and 22.3). The accompanying MSE verification also shows no sign of a topological obstruction: the rapid improvement slows at $m \approx 16$ –20.

Table 8. MNIST Step 3 (Exp-Mnist-Multi-300): AU and TwoNN ID of the VAE ($\beta = 4$, 300 epochs, mean $\pm$ std over 3 seeds). The AU growth slows beyond $m \approx \hat{d}_{\text{ID}}$, and the latent TwoNN converges to $\hat{d}_{\text{ID}} \approx 10$.

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 ± 0.0	3.98 ± 0.14	slowdown begins near $\hat{d}_{\text{ID}}$
8	8.0 ± 0.0	6.67 ± 0.15	
12	10.0 ± 0.0	7.41 ± 0.06	
16	11.7 ± 0.5	7.95 ± 0.21	
20	13.3 ± 0.5	8.37 ± 0.14	
32	15.7 ± 0.5	9.07 ± 0.10	
64	22.3 ± 0.5	10.03 ± 0.12	

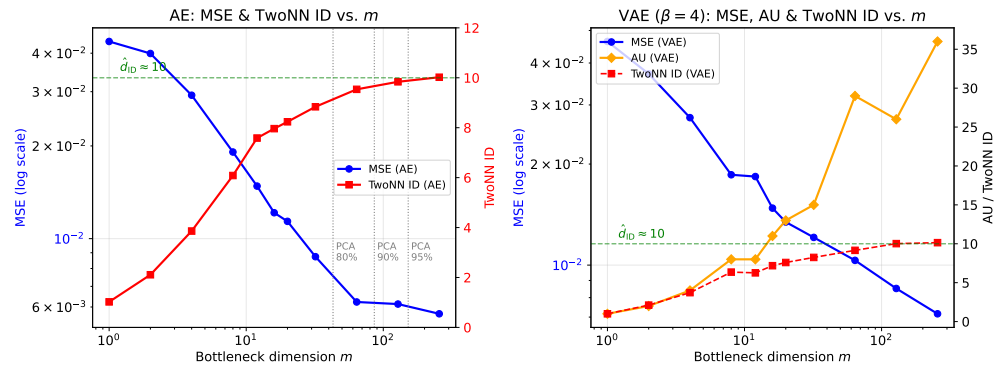


Figure 2. MNIST: m -dependence of the indicators (300 epochs). **Left (AE):** MSE (log scale, left axis) and TwoNN ID (right axis); vertical dotted lines mark the PCA 80/90/95% dimensions (43/86/152), far above the nonlinear estimate $\hat{d}_{ID} \approx 10$. **Right (VAE, $\beta = 4$):** MSE, AU, and TwoNN ID; the AU growth slows beyond $m \approx \hat{d}_{ID}$.

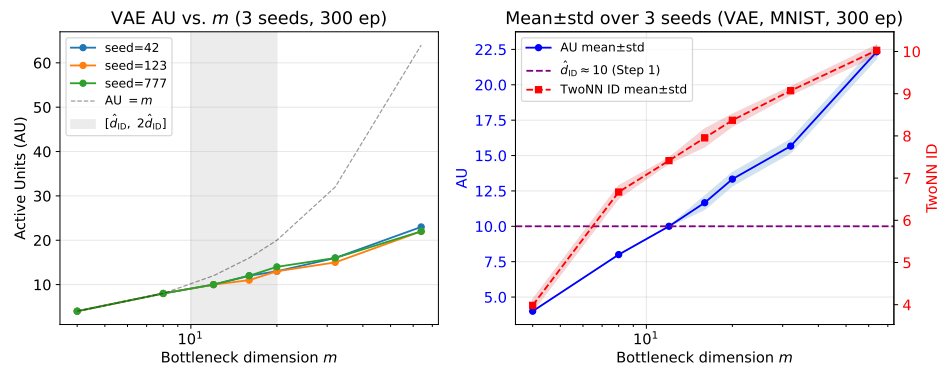


Figure 3. MNIST: multi-seed reproducibility of the AU verification signal (3 seeds, 300 epochs). Left: per-seed $AU(m)$ with the ID-guided search window $[\hat{d}_{ID}, 2\hat{d}_{ID}]$ shaded; right: mean $\pm$ std band with the Step-1 reference $\hat{d}_{ID} \approx 10$. AU values are highly reproducible ($\sigma \leq 0.5$). Change-point (knee) estimates are deliberately not shown here; they are exploratory only (Appendix B).

5.3.5. Fine-Grained View of the Slowdown Region and a Caution on Change-Point Detection

To inspect the onset of the slowdown more closely, we observed the AU curve on a dense grid with unit steps around $\hat{d}_{ID}$, $m \in \{8, 9, \dots, 16, 18, 20, 24, 28, 32, 40, 48, 64\}$ (17 points, 300 epochs, 3 seeds). $AU(m)$ stays on the diagonal ($AU = m$) up to $m = 8$ – 10 and departs from it at $m \geq 11$; applying a change-point detector based on first-difference ratios (threshold $\tau_{knee} = 0.25$; Appendix B) to this dense grid gives a change point of 10.3 ± 0.9 (per seed: 11, 9, 11), consistent with $\hat{d}_{ID} \approx 10$.

This agreement, however, carries two important qualifications. First, the dense-grid region was placed around the Step-1 TwoNN estimate, so the result is a **local curve summary conditional on a TwoNN-derived search window**, not an independent cross-validation. Second, applying the same detector to the sparse grid $\{4, 8, 12, 16, 20, 32, 64\}$ yields a change point of 43 ± 15 —**automatic change-point detection depends strongly on grid design and should not be operated as a named output of the protocol**. For this reason, Step 3 relies primarily on the robust read-out (multi-seed comparison of the AU values themselves)—whether the slowdown region is of the same order as $\hat{d}_{ID}$, and change-point detection is used only as an aid for grid design (detailed sensitivity analysis in Appendix B).

5.3.6. Fashion-MNIST

For Fashion-MNIST, Exp-Real-Geom (Section 5.5; 3 seeds, 300 epochs) gives $\hat{d}_{ID} \approx 10$: TwoNN converges to 9.9 – 10.1 at $m_{ref} = 5d$ – $10d$, where $d = 9$ is the sweep-grid design value

taken from the preliminary 100-epoch estimate (Appendix C). Step 1 is thus multi-seed for Fashion-MNIST; the VAE-side AU verification (Step 3), by contrast, remains a single-seed preliminary observation with the same tendency as MNIST (Appendix C), and multi-seed replication of the full protocol is future work.

5.4. Baseline Comparison: Search Cost and Downstream Performance

We quantify the practical value of ID-guided selection against a **full m -grid search** (Exp-Downstream). The evaluation metrics are the test classification accuracy of a linear probe (multiclass logistic regression) on the latent representations and the reconstruction MSE.

AE (reuse of the reference-AE sweep, 300 epochs, seed 42): as shown in the left half of Table 9, the linear-probe accuracy plateaus at $m = 8$ –12 (92.3% at $m = 12$) and does not improve up to $m = 256$ (91.8%; raw 784-dimensional pixels: 91.6%). **The best accuracy over the full grid (11 points), 92.3%, is attained inside the ID-guided search window [10, 24] (at $m = 12$),** and the other in-range points ($m = 16, 20$) are within 1.4 points of it.

VAE ($\beta = 4$, 300 epochs, 3 seeds): as shown in the right half of Table 9, the same picture holds on the VAE side: the linear-probe accuracy attains its best value over the grid $\{4, \dots, 64\}$, $89.1 \pm 0.4\%$, at $m = 12$ inside the search window, and is nearly flat for $m \geq 12$ (at the VAE-path verification point $m = 20 = 2\hat{d}_{\text{ID}}$ it is $88.2 \pm 0.4\%$, within 1 point of the best). The AU and MSE values in this table are retrained reproductions of Table 8 (Exp-Mnist-Multi-300) and agree with it—an additional reproducibility check.

Search cost: the full MNIST grid of this paper requires 11 values of $m \times 3$ seeds = 33 runs. The ID-guided path instead performs Step 1 (convergence check over $m_{\text{ref}} \in \{64, 128, 256\}$; 3 single-seed runs) and then either (a) the AE path: training the in-range grid points $\{12, 16, 20\}$ over 3 seeds (9 runs)—12 runs in total (**$\approx 64\%$ saved**); or (b) the VAE path: training the single point $m = 2\hat{d}_{\text{ID}} = 20$ over 3 seeds and verifying by AU (3 runs)—6 runs in total (**$\approx 82\%$ saved**). Under the conservative variant that also verifies at the MLE-shifted point $m = 32$ (Section 3.2.3), the VAE path is 9 runs in total (**$\approx 73\%$ saved**). Note that this accounting counts Step 1 as single-seed runs (this paper quantifies Step-1 uncertainty separately via subsampling CIs and estimator cross-checks); a symmetric accounting that repeats Step 1 over 3 seeds gives 18 runs for the AE path (**$\approx 45\%$ saved**) and 12 runs for the VAE path (**$\approx 64\%$ saved**). The retained performance is nearly identical to the grid-best as described above. This comparison is limited to the MNIST grid configuration and linear probe of this paper and does not establish a general reduction rate (Section 6.1).

Table 9. Exp-Downstream (MNIST): linear-probe test accuracy and MSE vs. m . Left: AE latents (300 epochs, single seed 42; reuse of the reference-AE sweep). Right: FC-VAE ($\beta = 4$) latents (300 epochs, mean $\pm$ std, 3 seeds). Bold m values lie in the ID-guided search window $[10, 24]$, which contains the best accuracy over the full grid; for the VAE, $m = 20$ is in addition the pre-specified verification point m_{ver} (Section 3.2.3). Raw-pixel baseline: 91.6%.

AE (single seed)			VAE ($\beta = 4$, 3 seeds)			
m	Probe acc. (%)	MSE	m	Probe acc. (%)	AU	MSE
1	60.0	0.0437	4	85.0 $\pm$ 0.5	4.0 $\pm$ 0.0	0.0282
2	60.0	0.0399	8	87.7 $\pm$ 0.1	8.0 $\pm$ 0.0	0.0182
4	81.7	0.0292	12	89.1 $\pm$ 0.4	10.0 $\pm$ 0.0	0.0154
8	91.7	0.0191	16	89.0 $\pm$ 0.2	11.7 $\pm$ 0.5	0.0138
12	92.3	0.0148	20	88.2 $\pm$ 0.4	13.3 $\pm$ 0.5	0.0126
16	92.0	0.0121	32	88.1 $\pm$ 0.2	15.7 $\pm$ 0.5	0.0112
20	90.9	0.0114	64	88.7 $\pm$ 0.3	22.3 $\pm$ 0.5	0.0089
32	91.2	0.0087				
64	91.8	0.0062				
128	92.2	0.0061				
256	91.8	0.0057				

5.5. Validity Condition of Step 1: What Makes Intrinsic-Dimension Estimation Stable?

Since Step 1 goes through the latents of a reference AE, the focus is whether the geometric distortion of the reference representation corrupts the estimate. In Exp-Real-Geom we swept four models with different isometry properties (plain AE / IsometricAE [17] / CAE [16] / DAE [15]) on MNIST and Fashion-MNIST over $m_{\text{ref}} \in \{d, 2d, 3d, 5d, 10d\}$ (d is the grid design value from preliminary estimates: 10 for MNIST, 9 for Fashion-MNIST), comparing TwoNN, MLE, κ , Trustworthiness, and kNN overlap (3 seeds, 300 epochs).

Table 10. Exp-Real-Geom (MNIST, mean $\pm$ std, 3 seeds): TwoNN through four reference models. Convergence is driven by the bottleneck margin $m_{\text{ref}}/\hat{d}_{\text{ID}}$, largely independently of the isometry (condition number κ).

Model	TwoNN ($m_{\text{ref}} = d$)	TwoNN ($m_{\text{ref}} = 5d$)	TwoNN ($m_{\text{ref}} = 10d$)	κ ($m_{\text{ref}} = 10d$)
AE	7.3 $\pm$ 0.4	10.0 $\pm$ 0.1	10.0 $\pm$ 0.1	4.1×10^7
IsometricAE	7.5 $\pm$ 0.4	10.1 $\pm$ 0.1	10.1 $\pm$ 0.1	7.8
CAE	7.2 $\pm$ 0.5	9.9 $\pm$ 0.1	10.0 $\pm$ 0.1	3.2×10^3
DAE	7.6 $\pm$ 0.4	10.0 $\pm$ 0.1	10.0 $\pm$ 0.1	6.3×10^3

The results (Table 10) are clear: (1) in all four models, TwoNN converges from the underestimate at $m_{\text{ref}} = d$ (≈ 7.3 – 7.6) to ≈ 10 at $m_{\text{ref}} \geq 5d$, and the inter-model differences (within ± 0.3) are an order of magnitude smaller than the effect of increasing m_{ref} (about 2.5 points); (2) IsometricAE improves κ from 10^7 to around 8, yet its effect on TwoNN stays within ± 0.2 . The same pattern was obtained on Fashion-MNIST. That is, **the dominant condition for Step-1 stability is not “improved isometry” but the “bottleneck margin $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ ”** (consistent with the δ_T -dominance implication of Proposition 3), justifying the design of Step 1 with a plain reference AE at $m_{\text{ref}} = 256 \gg \hat{d}_{\text{ID}}$.

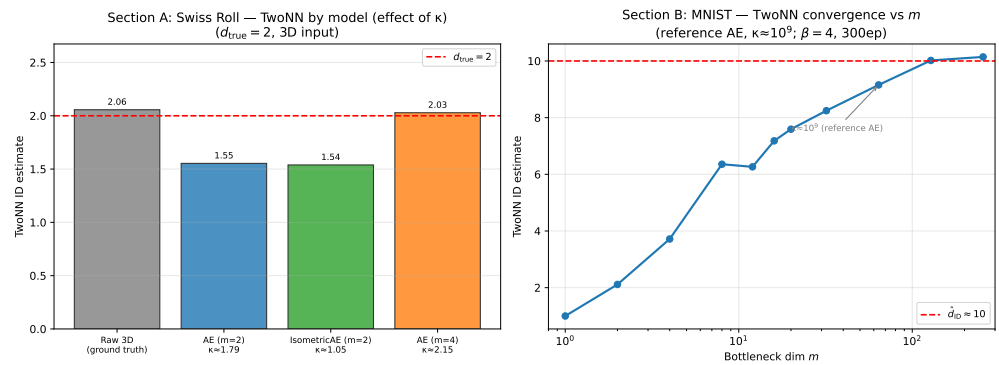


Figure 4. Step-1 stability: (left) on the Swiss Roll, TwoNN recovers once $m_{\text{ref}} > d_{\text{true}}$ regardless of isometry; (right) on MNIST, TwoNN stabilizes for $m_{\text{ref}} \geq 48 \approx 5\hat{d}_{\text{ID}}$ even where the condition number κ grows sharply.

5.6. Applicability Boundary: Conv-VAEs and Natural Images

This section is not a “report of failures” of the protocol but a quantification of the conditions under which the failure modes of Table 2 actually occur.

5.6.1. Conv-VAE: Truncation Failure Due to Insufficient β (3 Seeds)

We swept an MNIST Conv-VAE (2 Conv layers + 2 ConvTranspose layers, KL cyclic annealing with $\beta_{\text{max}} = 4$ [22], 150 epochs, 3 seeds) over $m \in \{4, 8, 16, 32, 64, 96, 128, 192, 256\}$ (Exp-Conv-M256). For $m \leq 64$, $\text{AU} = m$ (no truncation) persists in all seeds; AU falls below m for the first time at $m = 96$ ($\text{AU} = 90.7 \pm 1.2$), yet the activation rate remains as high as 94% at $m = 256$. Extending to $m \in \{384, 512\}$ (Exp-Conv-M512) still keeps $\text{AU} \approx m$ —that is, at $\beta_{\text{max}} = 4$ the truncation threshold satisfies $d^*(\beta) > 512$, and **the AU verification detects the state “ β too small relative to the decoder capacity” through the clear signal $\text{AU} = m$** (first row of Table 2).

From the viewpoint of the surrogate condition (Proposition 2), a powerful convolutional decoder makes the effective noise variance τ_{eff}^2 small (a back-computed calibration ≈ 0.30 for the FC-VAE and $\ll 0.30$ for the Conv-VAE; a heuristic extrapolation), lowering the threshold $\beta\tau_{\text{eff}}^2$ and inflating $d^*(\beta)$. This interpretation is consistent with the β -strengthening experiments: at $\beta_{\text{max}} = 10$, $\text{AU} = 93$ (36%) at $m = 256$; at $\beta_{\text{max}} = 20$, $\text{AU} = 66$ (26%)—a clear induced saturation (Exp-ConvV-HiBeta; **single-seed exploratory observation**, details in Appendix C). This is consistent with the qualitative prediction that increasing β moves the truncation to lower dimensions, identifying “strengthening β according to data and architecture” as a necessary condition for applying the protocol to Conv-VAEs.

Verdict (MNIST Conv-VAE): Step 1—usable (reusing $\hat{d}_{\text{ID}} \approx 10$ from the FC reference AE); Step 2—search window obtained ($m \geq 20$); Step 3—**fail** at $\beta_{\text{max}} = 4$ (activation ratio $0.94 > 0.9$ at $m = 256$; V1 violated). At $\beta_{\text{max}} \in \{10, 20\}$, V1 holds (activation ratios 0.36 and 0.26) but V2 remains violated since $\text{AU} = 93$ and $66 > 3\hat{d}_{\text{ID}} = 30$ (and the observation is single-seed exploratory); we judge this as “truncation induced, but order consistency with $\hat{d}_{\text{ID}}$ not yet achieved.”

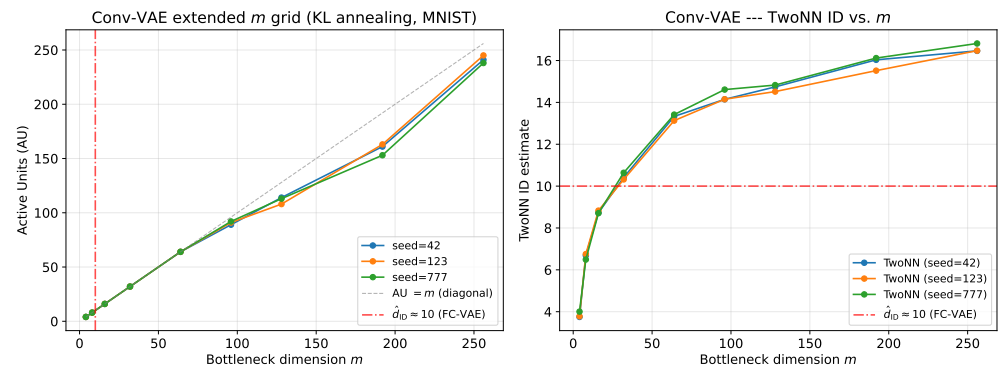


Figure 5. Boundary condition (Exp-Conv-M256): the MNIST Conv-VAE with $\beta_{\max} = 4$ keeps $AU = m$ (no truncation) up to $m = 64$ and stays at 94% activation even at $m = 256$ —the failure mode “ β too small relative to decoder power” detected by the AU signal.

5.6.2. Natural Images: Bounded Trust Region of Step 1 and Unmet Step 3 (3 Seeds)

For Conv-VAEs on CIFAR-10 and SVHN (3 Conv layers, $\beta_{\max} = 4$, 150 epochs, 3 seeds, $m \in \{16, 32, 64, 128, 256\}$), two observations follow (Table 11). First, **Step 1 works conditionally**: the latent TwoNN of CIFAR-10 at $m = 64 \approx 2\hat{d}_{ID}$ is 25.21 ± 0.41 , consistent with the raw-input estimate (25.20) and with Pope et al. [7] (≈ 26). Beyond $m \geq 128$, however, it inflates monotonically to 28.8–30.9—**on natural images the estimate is consistent with prior estimates only near $m_{\text{ref}} \approx 2\hat{d}_{ID}$, and “the larger the better” does not hold** (the true ID is unknown, so nothing stronger than consistency can be claimed; third failure mode of Table 2). The contrast with MNIST (stable even at $m_{\text{ref}} = 25\hat{d}_{ID}$) is attributed to latent-geometry distortion while all dimensions remain in use ($AU = m$, no truncation). For SVHN the discrepancy is even larger (latent 27.1 vs. raw 10.1 at $m = 256$). Second, **Step 3 is unmet**: no AU saturation appears for $m \leq 256$ (only $AU < m$ from $m \geq 128$ on SVHN); AU verification requires strengthened β , restricted decoder capacity, and exploration of larger m ranges (a preliminary observation of plateau-like behavior at $\beta_{\max} = 80$ with a thin decoder is reported in Appendix C).

Verdict (natural images): Step 1—usable only in the range $m_{\text{ref}} \approx 2\hat{d}_{ID}$ (CIFAR-10: 25.21 ± 0.41 , consistent with reported values; inflation beyond); Step 2—search window obtained conditionally (CIFAR-10: $m \geq 50$); Step 3—**unresolved** (CIFAR-10: activation ratio 1.0 at $m = 256$, V1 violated; SVHN: activation ratio 0.81 satisfies V1 but $AU = 206.7 \gg 3\hat{d}_{ID}$ violates V2).

Table 11. Natural images (Exp-Nat-Cifar / Exp-Nat-SVHN): Conv-VAE with KL cyclic annealing ($\beta_{\max} = 4$, 150 epochs, mean $\pm$ std, 3 seeds). For CIFAR-10, Step 1 agrees with Pope et al. [7] only near $m \approx 2\hat{d}_{ID}$ (for SVHN the latent-raw discrepancy is larger; see text); Step 3 (AU saturation) is not reached in the tested range.

m	CIFAR-10		SVHN	
	AU	TwoNN ID	AU	TwoNN ID
16	16.0 ± 0.0	13.12 ± 0.12	16.0 ± 0.0	11.30 ± 0.12
32	32.0 ± 0.0	19.26 ± 0.23	32.0 ± 0.0	15.08 ± 0.08
64	64.0 ± 0.0	25.21 ± 0.41	64.0 ± 0.0	19.97 ± 0.12
128	128.0 ± 0.0	28.83 ± 0.41	125.3 ± 0.5	24.60 ± 0.36
256	256.0 ± 0.0	30.91 ± 0.64	206.7 ± 1.2	27.08 ± 0.34
Raw-input TwoNN = 25.20, MLE = 26.56			Raw-input TwoNN = 10.09, MLE = 18.85	

6. Discussion

6.1. What Differs from Conventional Model-Selection Criteria?

The conventional means of choosing m in practice are m -sweeps of validation reconstruction error or ELBO, or downstream task performance. These tell us “the m at which performance plateaus,” but (i) they require sweeping the entire grid, and (ii) they do not distinguish the cause of the plateau (data manifold dimension, model capacity, or optimization). Our protocol does not replace them but **complements** them: Step 1 narrows the search window to the vicinity of $\hat{d}_{\text{ID}}$ before any sweep (reducing search cost), and the AU of Step 3 reports—independently of performance—how many dimensions are actually in use, directly visualizing the presence of redundant dimensions. Indeed, on MNIST the MSE verification, the AU verification, $\hat{d}_{\text{ID}}$, and the linear-probe accuracy all supported the same search window [10, 24] (Section 5.4), and on the MNIST grid used in this paper, choosing m inside the window reduced the number of training runs by roughly 60–80% relative to the full grid search while nearly retaining the downstream classification accuracy (no general reduction rate is claimed). This comparison is, however, limited to a linear probe on MNIST; systematic comparison with nonlinear downstream tasks and disentanglement metrics remains future work.

6.2. Stability vs. Validity: What Does $\hat{d}_{\text{ID}}$ Estimate?

A distinction maintained throughout this paper is that “TwoNN/MLE being **numerically stable** (low variance across seeds and m_{ref})” is different from “being **unbiased** with respect to the true intrinsic dimension.” The latent space of the reference AE is far from isometric (κ reaches 10^7 ; Table 10), and what TwoNN estimates is, strictly speaking, “the local degrees of freedom of the reference representation.” The MNIST value $\hat{d}_{\text{ID}} \approx 10$ –12 is supported by several pieces of indirect evidence—(i) convergence over multiple m_{ref} , (ii) TwoNN–MLE agreement, (iii) agreement across four reference models (Table 10), (iv) narrow subsampling CIs and order-of-magnitude agreement with estimators of different principles (DANCo, ESS; Table 7), and (v) consistency with reported values on CIFAR-10—but this is not a direct proof of unbiasedness. The protocol therefore operates $\hat{d}_{\text{ID}}$ as “a reference value giving the center of the search window” and closes the loop with the Step-3 verification. A remaining task is the calibration of $\hat{d}_{\text{ID}}$ against ground truth beyond synthetic data (a systematic evaluation of estimator bias).

6.3. Limits of AU Verification: Why Change-Point Detection Is Not a Named Output

An earlier version of this framework (the predecessor of this work) treated the AU slowdown point m^{knee} as one of its diagnostics; this paper removes it from the named outputs. The reasons, shown in Section 5.3.5, are that (i) the sparse-grid detection (43 ± 15) is dominated by the grid interval, (ii) stabilization requires a TwoNN-derived search window and hence is not independent validation, and (iii) the detection is also sensitive to the choice of detector (threshold vs. second difference) (Appendix B). The robust part of AU verification is the read-out “do the AU values themselves (multi-seed, $\sigma \leq 0.5$) slow down or saturate at the same order as $\hat{d}_{\text{ID}}$?”—the protocol is designed to use only this robust part in its decision rule (Equations (2)–(3), which involve no change-point detection). Nested validation, in which the search-window estimation and the change-point evaluation are performed on disjoint seeds or data subsets, remains future work.

6.4. Limitations and Future Work

- **Validation scope:** multi-seed validation of the full protocol is limited to synthetic manifolds and the MNIST AU verification (the MNIST Step-1 sweep is single-seed with separately quantified uncertainty; Fashion-MNIST is multi-seed for Step 1 only).

The recovery of Conv-VAE truncation by strengthened β (Exp-ConvV-HiBeta) is a single-seed exploratory observation and needs multi-seed replication.

- **Step 3 on natural images:** stable identification of AU saturation is unmet; dense sweeps of $\beta_{\max} \in [40, 120]$, decoder-capacity control, and improved seed stability are future work (Appendix C).
- **β dependence and decision thresholds:** the AU read-out depends on β (AU patterns confirmed stable for $\beta \in [2, 8]$; Appendix D). Actively exploiting the β dependence of $d^*(\beta)$ —a “spectral read-out by β sweep”—is an interesting extension. The thresholds of the decision rule (activation ratio 0.9, factor 3) are working defaults that separate the passing and failing cases of this paper by a wide margin; calibrating them on more diverse data is future work.
- **Scope of the theory:** Proposition 2 is a linear-Gaussian approximation, and its non-linear extrapolation rests on post-hoc consistency checks via τ_{eff}^2 . Deriving activation conditions in nonlinear settings is open.
- **Computational cost:** the multi-seed m -sweep of Step 3 takes hours to a day of GPU time. Step 1 shrinks the sweep range, but further lightening (e.g., early-epoch AU prediction) is future work.

7. Conclusions

To the question “can the bottleneck dimension m of AEs/VAEs be determined from the manifold dimension of the input data,” this paper gave an answer in the form of a **conditional, diagnostic ID-guided protocol**: (S1) intrinsic-dimension estimation through reference representations with $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$, (S2) setting m as $\hat{d}_{\text{ID}}$ plus a topological margin (AE: $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$; VAE: $m \geq 2\hat{d}_{\text{ID}}$), and (S3) post-training verification by AU and MSE equipped with the pass/fail rule V1/V2. The claim takes a conditional form rather than that of a general solution: **$\hat{d}_{\text{ID}}$ provides a search window, and AU diagnoses whether the VAE actually truncated the surplus dimensions under a suitable β and architecture**. Theoretically, the topological lower bound and the water-filling surrogate activation condition $\lambda_j > \beta\tau^2$ provide the rationale for each protocol step; experimentally, the full protocol operated as designed on synthetic manifolds with known ground truth ($\text{AU}_{\text{sat}} = d_{\text{true}}$ or d_{emb} under sufficient margin— $m \geq 5$ on the Torus—with the boundary over-pruning at $m \leq 4$ detected by the MSE cross-check; 3–5 seeds, $\sigma = 0$) and on fully connected VAEs on MNIST ($\hat{d}_{\text{ID}} = 10.03 \pm 0.12$ consistent with the AU slowdown region; 3 seeds), with the ID-guided window reducing the training runs of the full grid search used in this paper by roughly 60–80% while retaining downstream classification accuracy. At the same time we quantified the applicability boundaries: intrinsic-dimension estimation is stable only with bottleneck margin $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ (isometry improvements are secondary; four-model comparison), inflates beyond $m_{\text{ref}} \approx 2\hat{d}_{\text{ID}}$ on natural images, and for Conv-VAEs an insufficient β is detected as a V1 violation ($\text{AU} = m$); strengthening β restores truncation but order consistency with $\hat{d}_{\text{ID}}$ (V2) remains unmet (a single-seed exploratory observation). The main future tasks are multi-seed stable identification of Step 3 on natural images and Conv-VAEs, nested validation of change-point detection, and bias calibration of $\hat{d}_{\text{ID}}$.

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Data Availability Statement: MNIST [27], Fashion-MNIST [28], CIFAR-10 [25], and SVHN [26] are publicly available. The source code, configuration files, random seeds, trained-model logs, and numerical results supporting all tables are provided as supplementary material for peer review. Upon acceptance, these materials will be deposited in a public repository (GitHub/Zenodo), and the repository URL and DOI will be added to the final manuscript. The repository will include the scripts required to reproduce the AU, TwoNN/MLE, reconstruction-error, and grid-sensitivity analyses reported in this paper.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AE	Autoencoder
VAE	Variational Autoencoder
AU	Active Units
ID	Intrinsic Dimension
TwoNN	Two-Nearest-Neighbor (ID estimator)
MLE	Maximum Likelihood Estimation
MSE	Mean Squared Error
PCA	Principal Component Analysis
KL	Kullback–Leibler (divergence)
FC	Fully Connected
Conv	Convolutional
CAE	Contractive Autoencoder
DAE	Denoising Autoencoder
CKA	Centered Kernel Alignment

Appendix A. Proofs and Derivations

Appendix A.1. Derivation Sketch of Proposition 2 (Surrogate Activation Condition)

Consider a β -VAE with linear encoder $f(x) = W_e x$, linear decoder $g(z) = W_d z$, prior $p(z) = \mathcal{N}(0, I_m)$, and likelihood $p(x|z) = \mathcal{N}(W_d z, \tau^2 I_N)$. Taking the input in its PCA basis with covariance $\Sigma = \text{diag}(\lambda_1, \dots, \lambda_N)$ ($\lambda_1 \geq \lambda_2 \geq \dots$), the β -ELBO separates across eigendirections in the linear-Gaussian model. Let $R_j \geq 0$ be the rate (KL contribution) allocated to direction j and $D_j(R_j)$ the corresponding distortion reduction; the Gaussian rate–distortion function gives $D_j(R_j) = \lambda_j e^{-2R_j}$ (nats), so the relevant term of the β -ELBO is

$$\mathcal{L}_j(R_j) = -\frac{1}{2\tau^2} \lambda_j e^{-2R_j} - \beta R_j + \text{const.} \quad (\text{A1})$$

Optimization over R_j is a concave program, and the KKT condition $\partial \mathcal{L}_j / \partial R_j = 0$ yields $R_j^* = \max(0, \frac{1}{2} \log(\lambda_j / \beta \tau^2))$. The activation condition (Equation (4)) is $R_j^* > 0 \iff \lambda_j > \beta \tau^2$. This coincides with Shannon’s water-filling theorem [23] with water level $\beta \tau^2$, and reduces to the known PPCA result [12] for $\beta = 1$. With m available coordinates the optimum activates the top $\min(m, d^*(\beta))$ directions, giving claims (2) and (3).

Content of the approximation: the derivation replaces the expected KL term $\mathbb{E}_x[\text{KL}(q_\phi(z|x) \| p(z))]$ by the mutual information $I(x; z)$ (the rate). Their difference is the aggregated-posterior mismatch $\text{KL}(q_\phi(z) \| p(z)) \geq 0$, and the proposition holds only where this term is negligible. The gradient dynamics of trained nonlinear VAEs are not treated, so the implications for nonlinear settings are qualitative.

Appendix A.2. Proof Sketch of Proposition 3 (Conditional Stability of TwoNN)

By the bi-Lipschitz condition, each distance changes by a factor within $(1 \mp \varepsilon)^2$. The TwoNN statistic r_2/r_1 is a ratio of distances, so uniform scaling cancels and only the

non-isometric component ($\varepsilon \neq 0$) distorts the ratio. Trustworthiness $\geq 1 - \delta_T$ guarantees rank preservation of the two nearest neighbors with high probability, probabilistically suppressing large variations of r_2/r_1 due to rank reversals. Under both conditions, the change (EMD) of the empirical distribution of r_2/r_1 is bounded by a continuous function $C(\varepsilon, \delta_T)$, and so is the deviation of the TwoNN estimate ($C = 0$ at $\varepsilon = \delta_T = 0$). This is a finite-sample, finite-dimension approximation; rigorous constants depend on the specific setting.

Appendix A.3. Other Interpretive Propositions

Pullback metric and local isometry: the decoder g is locally isometric if and only if $G(z) = J_g(z)^\top J_g(z) = I_m$. The condition number $\kappa = \sigma_{\max}(J_g)/\sigma_{\min}(J_g)$ measures anisotropy; $\kappa = 1$ implies only isotropy, not isometry. Moreover, κ is a local metric diagnostic and does not fully characterize the neighbor-rank distortions that drive TwoNN. In this paper κ is used as an auxiliary indicator of distortion.

Effective rank of the decoder Jacobian: when an AE of sufficient capacity reconstructs a manifold $\mathcal{M}$ exactly, the number of significant singular values of $J_g(z)$ equals $\min(m, d_{\text{emb}})$ (an idealization neglecting finite capacity and optimization error).

Appendix B. Exploratory Analysis of AU Change-Point Detection

This appendix collects the sensitivity data behind the decision (Section 6.3) **not** to make the automatic detection of a change point (knee) m^{knee} on the AU curve a named protocol output.

Appendix B.1. Definition

On a grid $\mathcal{M} = \{m_1 < \dots < m_K\}$, define the AU increase rate $s(m_i) = [\text{AU}(m_i) - \text{AU}(m_{i-1})]/(m_i - m_{i-1})$ and the normalized rate $\rho(m_i) = s(m_i)/\max_j s(m_j)$, and set

$$m^{\text{knee}} = \min\{m_i \in \mathcal{M} : \rho(m_i) < \tau_{\text{knee}}\}, \quad \tau_{\text{knee}} = 0.25 \quad (\text{A2})$$

(first-difference ratio method). An alternative is the second-difference (d2) method, which takes the point of maximum absolute discrete curvature of $\text{AU}(m)$.

Appendix B.2. Grid Sensitivity

In an idealized monotone-saturation model (AU saturates stepwise at $d^*(\beta)$ without noise), $|m^{\text{knee}}(\mathcal{M}) - d^*(\beta)| \leq \max_i (m_{i+1} - m_i)$ holds (the detection error is bounded by the maximum grid interval). On the actual MNIST FC-VAE ($\beta = 4$, 300 epochs, 3 seeds), stochastic fluctuations of the empirical AU curve produce variation beyond this bound (Table A1).

Table A1. Grid sensitivity of the change-point detector on the MNIST FC-VAE ($\tau_{\text{knee}} = 0.25$, 3 seeds). The sparse-grid value is dominated by the grid interval; the dense grid is placed around the TwoNN estimate and is therefore a TwoNN-conditioned curve summary, not independent validation.

Condition	Max. interval	seed 42	seed 123	seed 777	Mean $\pm$ std
Sparse grid, 150 ep	32	64	32	64	53 $\pm$ 15
Sparse grid, 300 ep	32	64	32	32	43 $\pm$ 15
Dense grid, 300 ep	16 (step 1 in 8–16)	11	9	11	10.3 $\pm$ 0.9

Extending the epochs (150 $\rightarrow$ 300) has little effect; the dominant factors are grid design and seed, not insufficient convergence. A single-seed preliminary observation on a different grid also gave $m^{\text{knee}} = 12$ (Exp-Mnist-Base), making the grid/initialization dependence explicit.

Appendix B.3. Detector and Threshold Sensitivity

Table A2 compares the τ -method and the d2 method on the dense-grid data. The d2 method is sensitive to AU fluctuations in the transition region (e.g., $8 \rightarrow 10 \rightarrow 9 \rightarrow 11 \rightarrow 10 \rightarrow 11 \rightarrow 13$) and deviates strongly for seed 123 ($m^{\text{knee}} = 28$). The value $\tau_{\text{knee}} = 0.25$ is a working default of this study; on Fashion-MNIST a boundary sensitivity (the detected value changes around τ) has been confirmed—we claim no general meaning for τ_{knee} .

Table A2. Detector sensitivity on the dense grid (300 ep, 3 seeds).

Method	seed 42	seed 123	seed 777	Mean $\pm$ std
τ -method ($\tau = 0.25$)	11	9	11	10.3 ± 0.9
Second difference (d2)	15	28	12	18.3 ± 6.8
Reference: $\hat{d}_{\text{ID}} \approx 10$ (TwoNN, 3-seed mean)				

Appendix B.4. Conclusion

Change-point detection is sensitive to grid, seed, detector, and threshold, and its stabilization requires the circular condition of a TwoNN-derived search window. The protocol therefore confines change points to an aid for grid design (where to sample densely around $\hat{d}_{\text{ID}}$), and bases its verification verdicts on multi-seed comparison of the AU values themselves. Nested validation separating window estimation from change-point evaluation remains future work.

Appendix C. Single-Seed and Exploratory Experiments

All experiments in this appendix are **single-seed (42) exploratory observations** and are not used as direct evidence for the main-text claims.

Appendix C.1. High- β Induction for the Conv-VAE (Exp-ConvV-HiBeta)

Table A3 shows the effect of strengthening β_{max} for the MNIST Conv-VAE (KL cyclic annealing, 150 epochs, $m \in \{16, 32, 64, 96, 128, 192, 256\}$). At $\beta_{\text{max}} = 10$ the activation rate drops to 36% at $m = 256$ and the change-point detector (τ -method) returns ≈ 128 ; at $\beta_{\text{max}} = 20$ it moves further down (≈ 96). This is consistent with the qualitative prediction of the surrogate condition that increasing β moves the truncation to lower dimensions.

Table A3. Exp-ConvV-HiBeta (single seed 42, exploratory): AU of the MNIST Conv-VAE under strengthened β_{max} (activation rate in parentheses).

m	$\beta_{\text{max}} = 4$	$\beta_{\text{max}} = 10$	$\beta_{\text{max}} = 20$
32	32 (100%)	32 (100%)	27 (84%)
64	64 (100%)	54 (84%)	42 (66%)
128	114 (89%)	69 (54%)	53 (41%)
256	241 (94%)	93 (36%)	66 (26%)

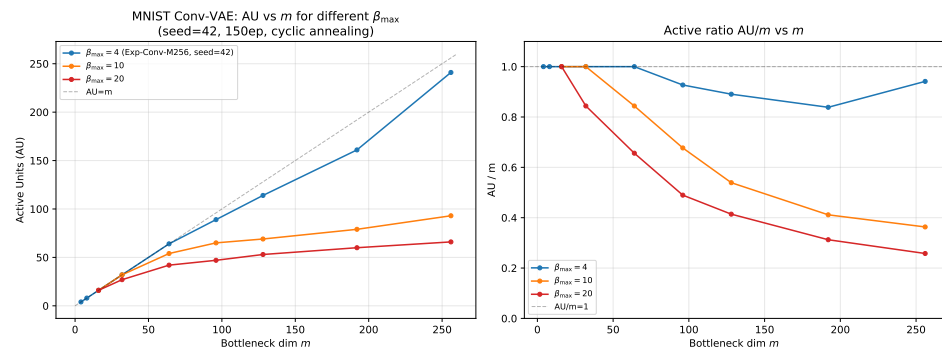


Figure A1. Exp-ConvV-HiBeta (single seed, exploratory): increasing $\beta_{\max}$ shifts the AU truncation to lower dimensions, consistent with the surrogate activation condition.

Appendix C.2. Search for AU Plateaus on Natural Images (Exp-NatImg-Sweep)

In a sweep of the CIFAR-10 Conv-VAE combining $m \in [32, 1024]$, $\beta_{\max} \in \{10, 20, 40, 80\}$, and decoder capacities (Standard/Deep/Thin), plateau-like behavior of AU around ≈ 24 was observed over $m \in [512, 1024]$ under the condition $\beta_{\max} = 80$ with a thin decoder (preliminary evidence). Across all conditions the latent TwoNN remained stable at 24.7–25.4, consistent with Pope et al. [7]. Stable identification including seed and grid stability is unmet and remains future work.

Appendix C.3. Preliminary VAE Observation on Fashion-MNIST (Exp-Fashion)

For the Fashion-MNIST VAE ($\beta = 4$, 100 epochs, single seed), the AE side TwoNN converges to ≈ 9.4 for $m_{\text{ref}} \geq 32$, and an AU slowdown region of the same order as $\hat{d}_{\text{ID}} \approx 9$ –10 was observed (same tendency as MNIST). Multi-seed replication is future work (the multi-seed Step-1 values come from Exp-Real-Geom; Section 5.5).

Appendix C.4. AU_{sat} on Topological Manifolds (Exp-Syn-Top)

On the Torus and the Möbius band (both $d_{\text{ID}} = 2$, $d_{\text{emb}} = 3$) we observed $AU_{\text{sat}} = 3 = d_{\text{emb}}$, and on the circle S^1 ($d_{\text{emb}} = 2$) $AU_{\text{sat}} = 2$ (for $m \geq 6$, conditional). The AU_{sat} of the Swiss Roll depends on preprocessing (2 with standardization, 3 otherwise), so the main evidence for the $AU_{\text{sat}} = d_{\text{emb}}$ correspondence is limited to the Torus and Möbius band. This is a conditional incidental observation with no claim of theoretical necessity.

Appendix D. Sensitivity Analyses and Experiment Overview

AU threshold δ : AU values are essentially unchanged for $\delta \in [10^{-3}, 10^{-1}]$ (MNIST FC-VAE, 300 epochs).

β sensitivity: in preliminary experiments (50 epochs, $\beta \in \{1, 2, 4, 8\}$), $\beta = 1$ gives $AU \approx m$ (under-regularization) and $\beta = 8$ shows signs of over-regularization; AU patterns are stable for $\beta \in [2, 8]$. The default $\beta = 4$ of this paper is the center of this stable range.

k sensitivity of MLE: MLE = 11.5–12.4 (MNIST) for $k \in \{5, 10, 20\}$, stable.

Table A4. Experiment overview. Main-text VAE verification results used for the principal verdicts use 3 seeds (42, 123, 777); single-seed sweeps are marked in the Seeds column; appendix experiments marked “single” use seed 42 only.

Experiment	Data / model	Epochs	Seeds	Section
Exp-Syn-1 / Exp-Syn-AU	Swiss Roll, Torus / AE, VAE	200–300	3	Section 5.2
Exp-Syn-Verify	Torus, Swiss Roll / VAE ($m = 3-8$)	200	5	Section 5.2
Exp-Mnist-Base	MNIST / AE (m_{ref} sweep)	300	1 (sweep)	Section 5.3
Exp-Mnist-Multi-300	MNIST / FC-VAE	300	3	Section 5.3
Exp-Mnist-Dense-300	MNIST / FC-VAE (dense grid)	300	3	Section 5.3.5
Exp-Boot-CI	MNIST / ref.-AE latents (re-analysis)	—	subsamp.	Section 5.3.2
Exp-Downstream	MNIST / AE probe, FC-VAE probe	300	1 (AE), 3 (VAE)	Section 5.4
Exp-Real-Geom	MNIST, F-MNIST / AE, IsoAE, CAE, DAE	300	3	Section 5.5
Exp-Conv-M256 / M512	MNIST / Conv-VAE	150	3	Section 5.6
Exp-Nat-Cifar / Exp-Nat-SVHN	CIFAR-10, SVHN / Conv-VAE	150	3	Section 5.6
Exp-ConvV-HiBeta	MNIST / Conv-VAE (β_{max} up)	150	single	App. C
Exp-NatImg-Sweep	CIFAR-10 / Conv-VAE	300	partial	App. C
Exp-Fashion	F-MNIST / AE, VAE	100	single	App. C
Exp-Syn-Top	Torus, Möbius, S^1 / VAE	300	single	App. C

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